MODULE – 1
INTRODUCTION TO MACHINE LEARNING

MODULE 1 - SYLLABUS

- Machine Learning terminology
- Types of Machine Learning
- Issues in Machine Learning
- Application of Machine Learning
- Steps in developing ML application
- How to choose the right algorithm
- Hypothesis Testing

Introduction

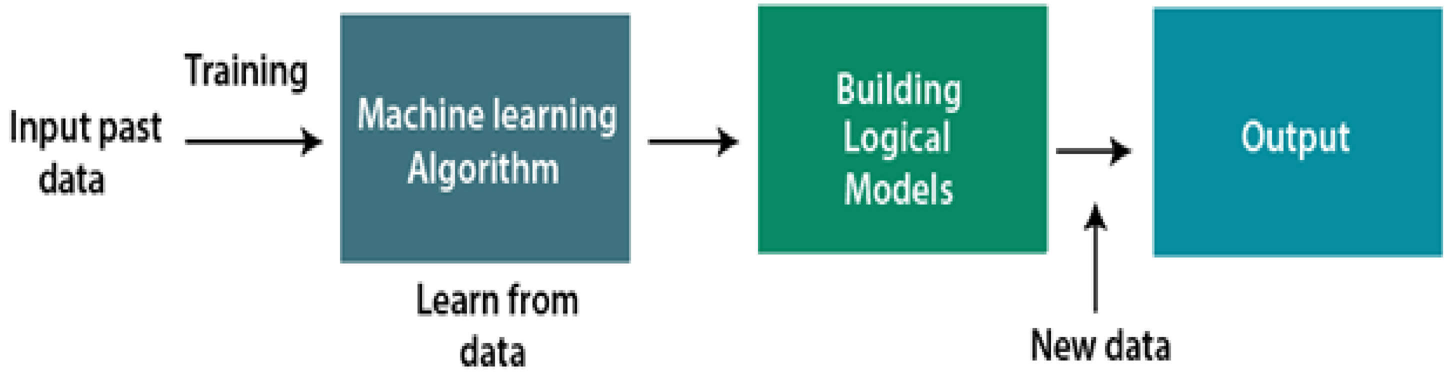
- Machine Learning is said as a subset of Artificial Intelligence (AI)
- It is mainly concerned with the development of algorithms which allow a computer to learn from the data and past experiences on their own.
- It improve performance from experiences, and predict things without being explicitly programmed.

Introduction

- Using sample historical data (training data), ML algorithms build a mathematical model that helps in making predictions
- It brings computer science and statistics together for creating predictive models
- The more we will provide the information, the higher will be the performance.

Working of Machine Learning

- Learn
- Make decisions
- A ML system **learns from historical data**, **builds the prediction models**, **and whenever it receives new data, predicts the output for it**



- The accuracy of predicted output depends upon the amount of data.

Traditional Programming Vs. Machine Learning

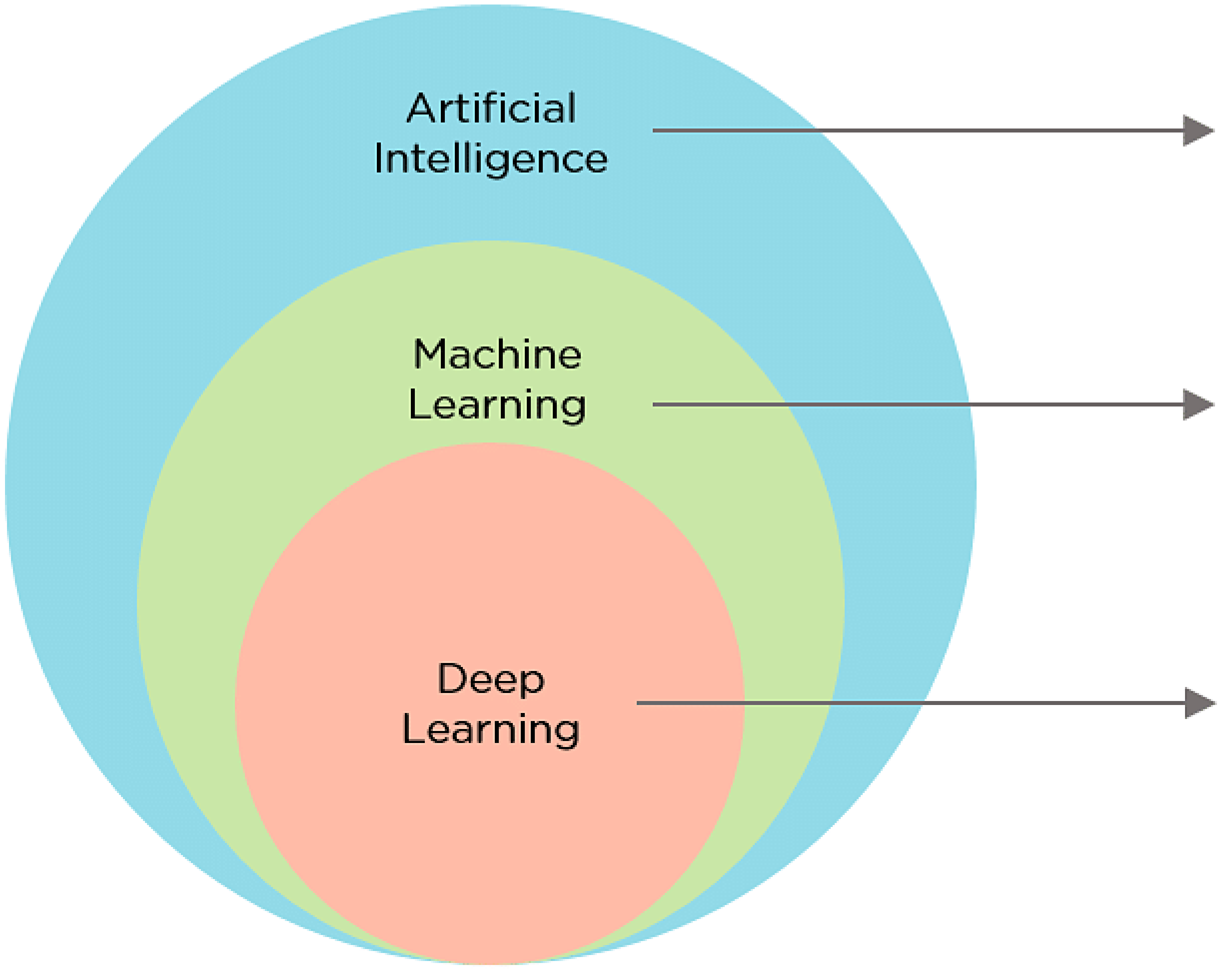
Need of Machine Learning

- It is capable of doing tasks that are too complex for a person.
- Machine learning is widely used in many industries, including healthcare, finance, and e-commerce
- Using Machine Learning, we can save both time and money.
- Machine learning is an important tool for data analysis and visualization.
- **Use Case:**
  - Self-driving cars,
  - Cyber fraud detection,
  - Friend suggestion by facebook,
  - Facial recognition systems,etc.

Advantages of Machine Learning

- Rapid increment in the production of data
- Solving complex problems, which are difficult for a human
- Decision making in various sector including finance
- Finding hidden patterns and extracting useful information from data

AI ,ML, DL



Ability of a machine to imitate intelligent human behavior

Application of AI that allows a system to automatically learn and improve from experience

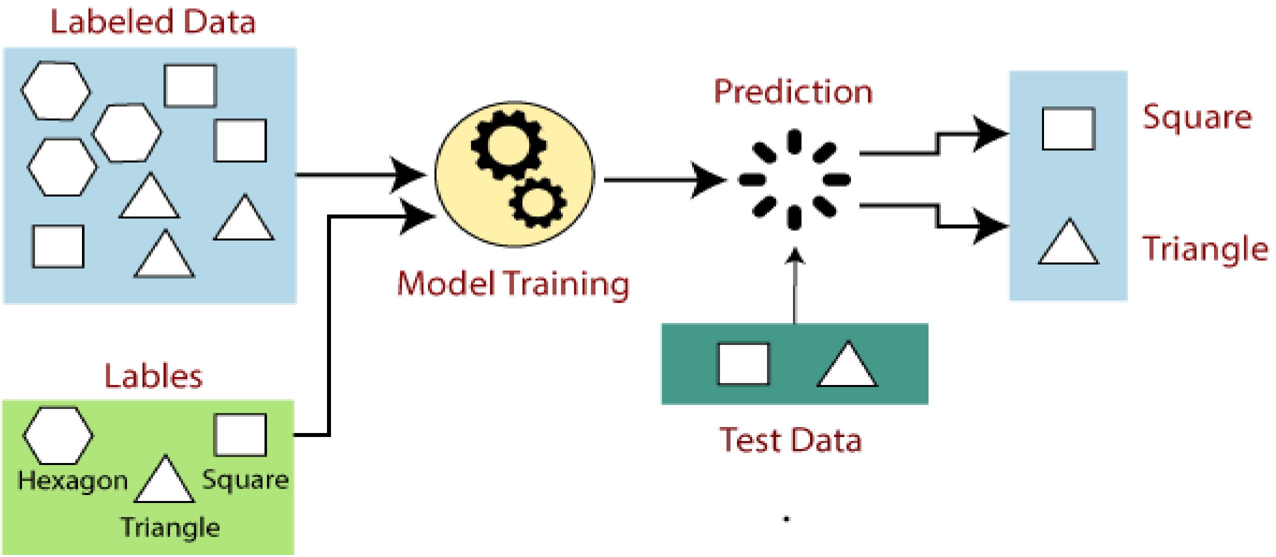
Application of Machine Learning that uses complex algorithms and deep neural nets to train a model

Types of Machine Learning

1. Supervised Learning
2. Unsupervised Learning
3. Reinforcement Learning

Supervised learning

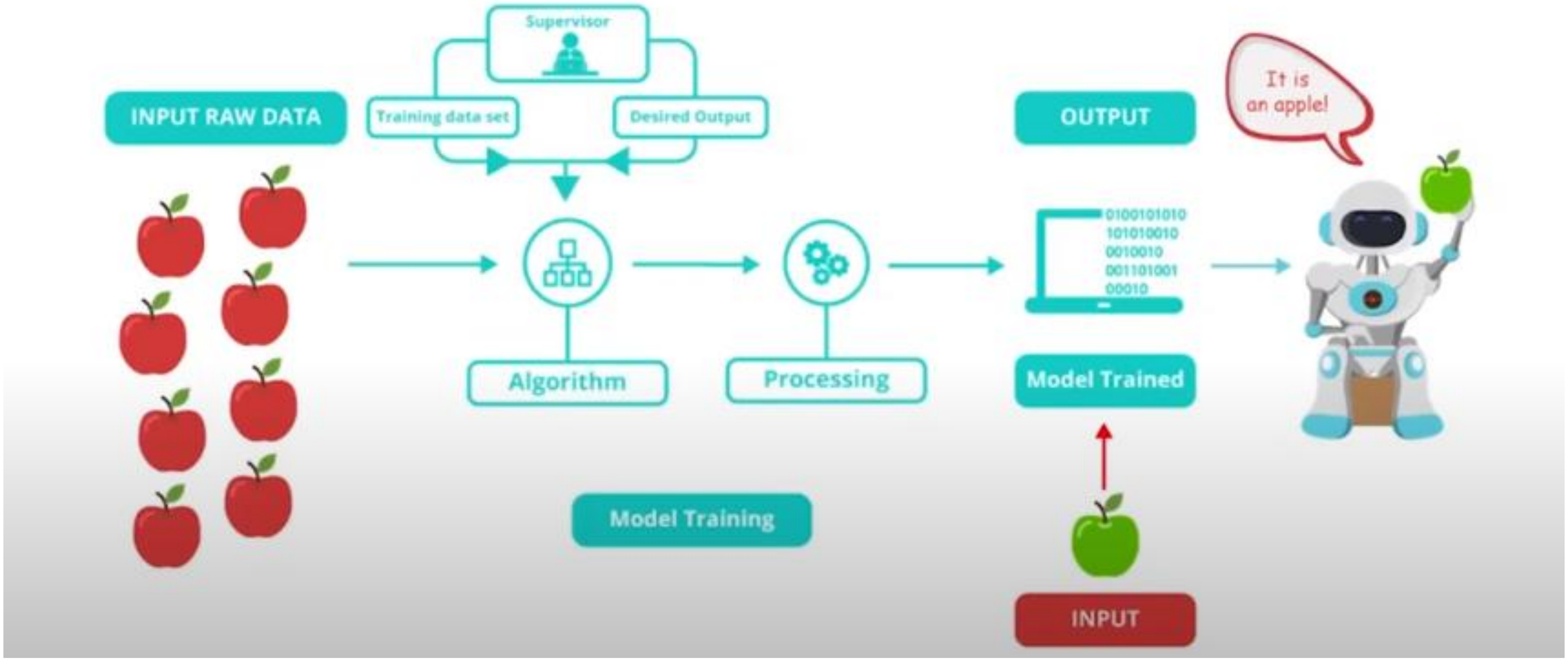
- Supervised learning is the types of machine learning.
- Machines are trained using well "**labelled**" training data, and on basis of that data, machines predict the output.
- Training data is provided to the machines work as the **supervisor** that teaches the machines to predict the output correctly.



Supervised learning

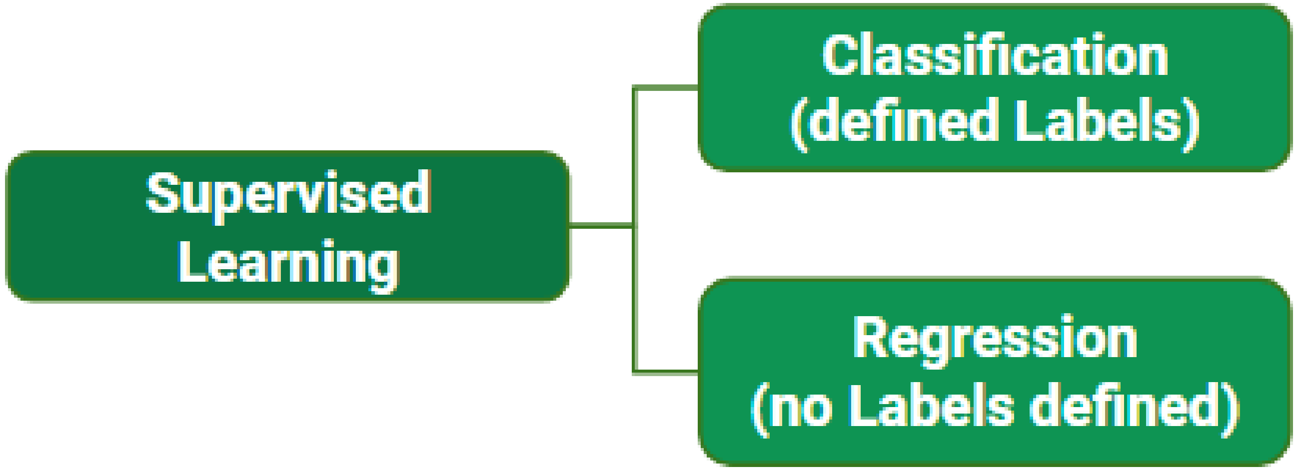
- Supervised learning is a process of providing input data as well as correct output data to the machine learning model.
- The aim of a supervised learning algorithm is to **find a mapping function to map the input variable(x) with the output variable(y)**.
- You are given reviews of few netflix series marked as positive, negative and neutral. Classifying reviews of a new netflix series is an example of Supervised learning

Supervised learning



Types of Supervised learning

- Supervised learning can be applied to two main types of problems:
- Classification: Where the output is a categorical variable (e.g., spam vs. non-spam emails, yes vs. no).
- Regression: Where the output is a continuous variable (e.g., predicting house prices, stock prices).



Types of Supervised learning

Classification:

- Classification algorithms are used to solve the classification problems in which the output variable is categorical, such as **"Yes" or No, Pass or Fail, etc.**
- The classification algorithms predict the categories present in the dataset.
- Some real-world examples of classification algorithms are **Spam Detection, Email filtering, etc.**

Types of Classification

Classification Types

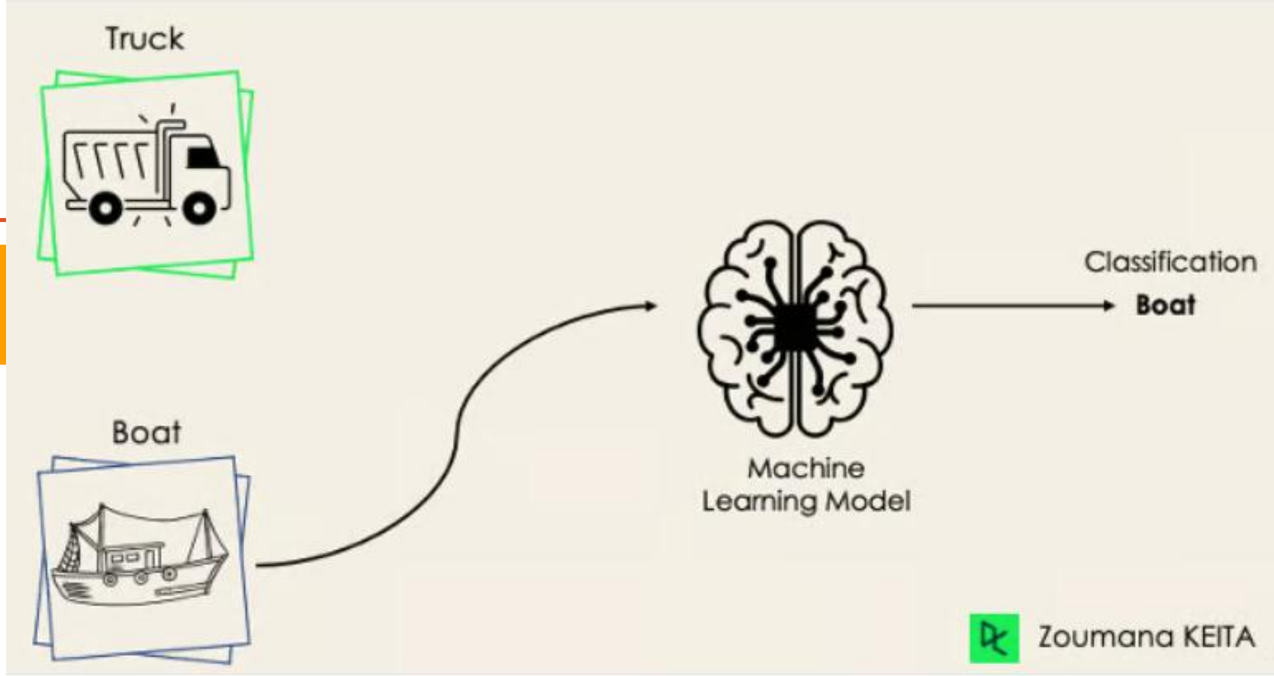
1. Binary Classification
2. Multi-Class Classification
3. Multi-Label Classification
4. Imbalanced Classification

1. Binary Classification:

- Binary classification refers to those classification tasks that have two class labels. The training data in such a situation is labeled in a binary format: true and false; positive and negative; O and 1; spam and not spam, etc. depending on the problem being tackled. For instance, we might want to detect whether a given image is a truck or a boat.
- Examples include:
  - Email spam detection (spam or not).
  - Conversion prediction (buy or not).

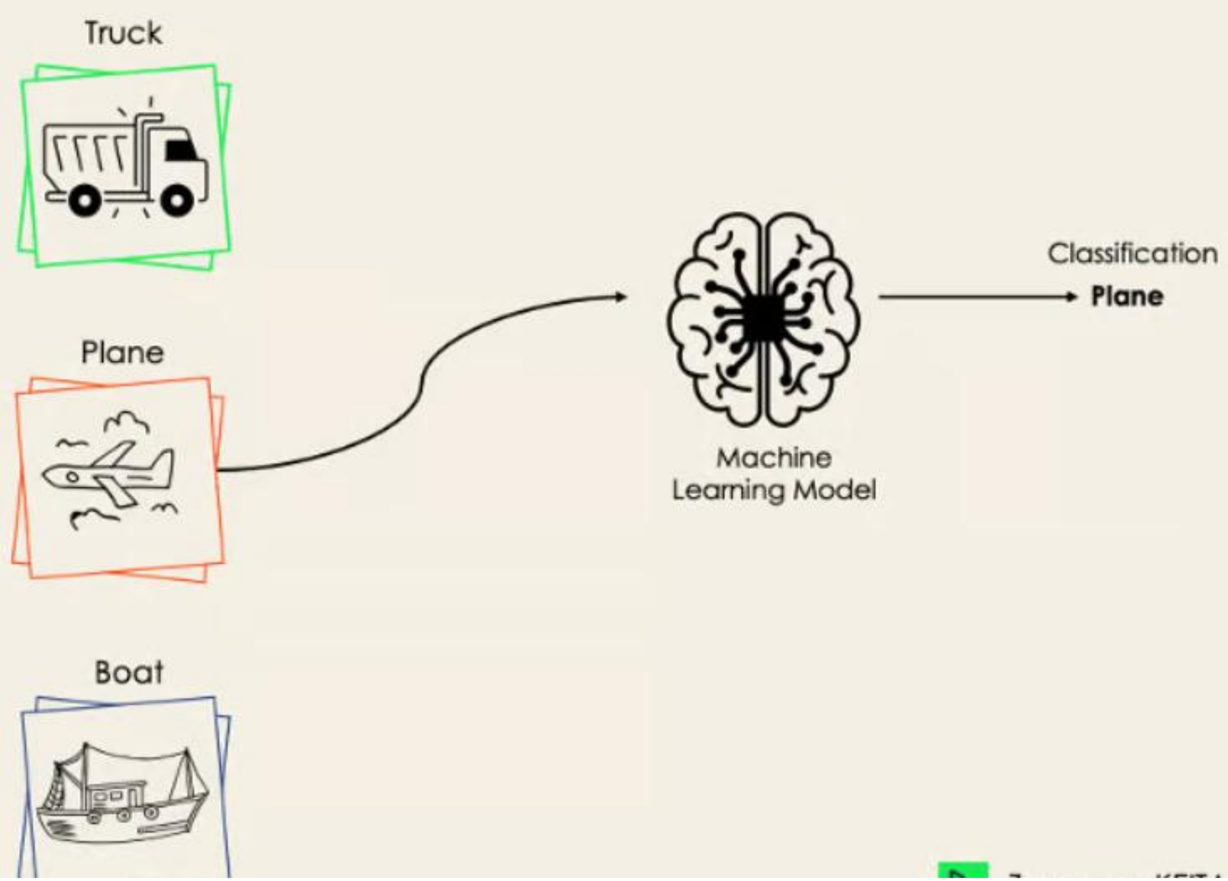
Popular algorithms :

- Logistic Regression
- k-Nearest Neighbors
- Decision Trees
- Support Vector Machine
- Naive Bayes



2. Multi-Class Classification

- The multi-class classification, on the other hand, has at least two mutually exclusive class labels, where the goal is to predict to which class a given input example belongs to. In the following case, the model correctly classified the image to be a plane.
- Examples include:
  - Face classification.
  - Plant species classification.
  - Optical character recognition.

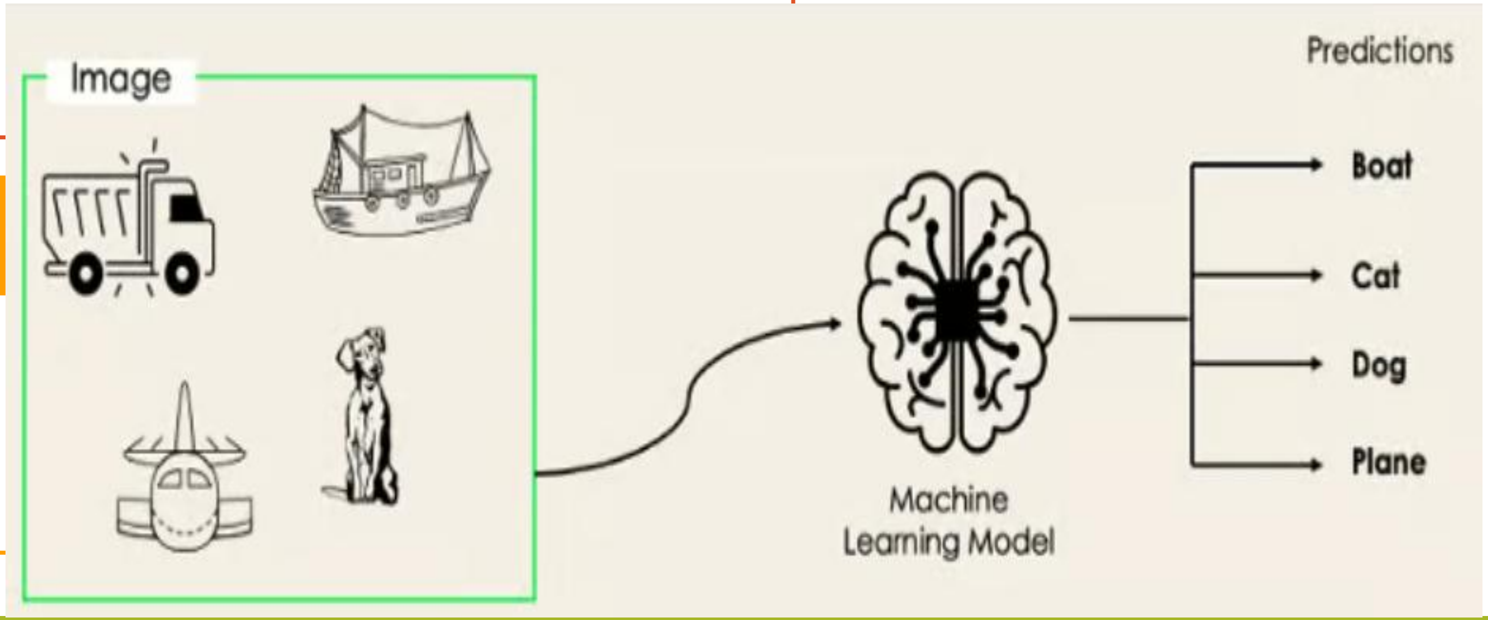


Popular algorithms :

- k-Nearest Neighbors.
- Decision Trees.
- Naive Bayes.
- Random Forest.
- Gradient Boosting.

3. Multi-Label Classification

- In multi-label classification tasks, we try to predict 0 or more classes for each input example. In this case, there is no mutual exclusion because the input example can have more than one label.
- Such a scenario can be observed in different domains, such as auto-tagging in Natural Language Processing, where a given text can contain multiple topics. Similarly to computer vision, an image can contain multiple objects, as illustrated below: the model predicted that the image contains: a plane, a boat, a truck, and a dog.

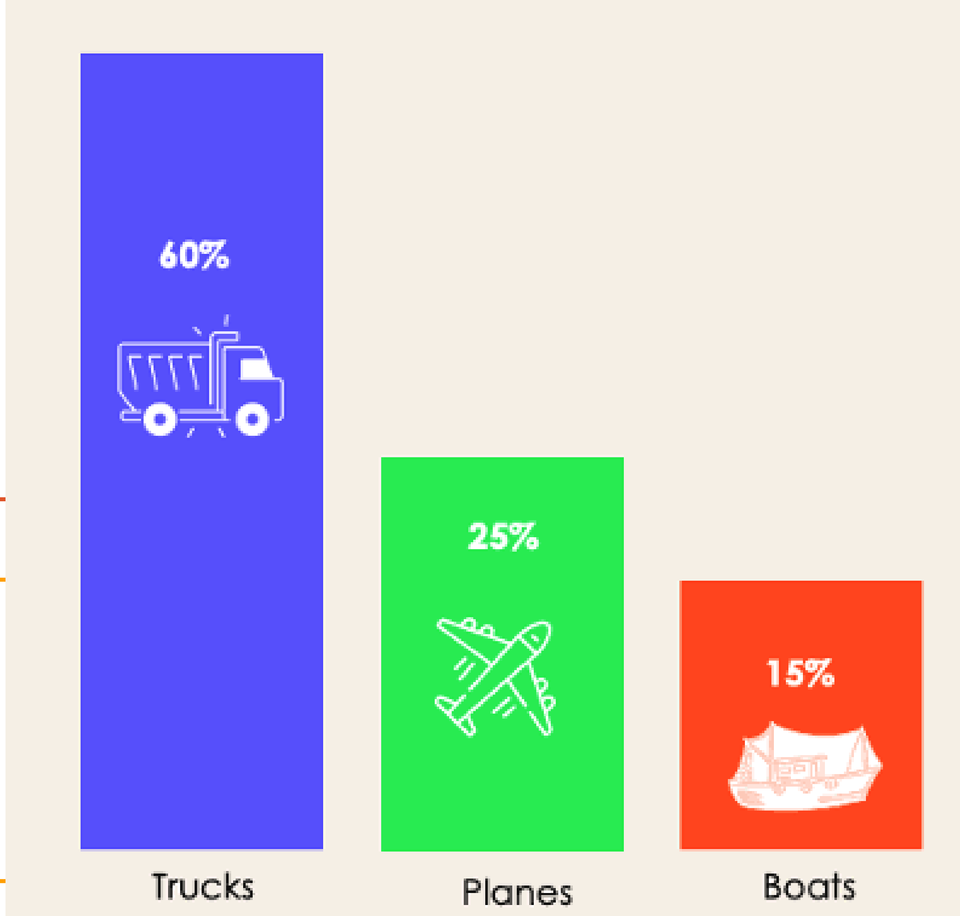


Popular algorithms :

- Multi-label Decision Trees
- Multi-label Random Forests
- Multi-label Gradient Boosting

4. Imbalanced Classification

- For the imbalanced classification, the number of examples is unevenly distributed in each class, meaning that we can have more of one class than the others in the training data. Let's consider the following 3-class classification scenario where the training data contains: 60% of trucks, 25% of planes, and 15% of boats.
- Examples include:
  - Fraud detection.
  - Outlier detection.
  - Medical diagnostic tests.



Popular algorithms :

- Cost-sensitive Logistic Regression.
- Cost-sensitive Decision Trees.
- Cost-sensitive Support Vector Machines.

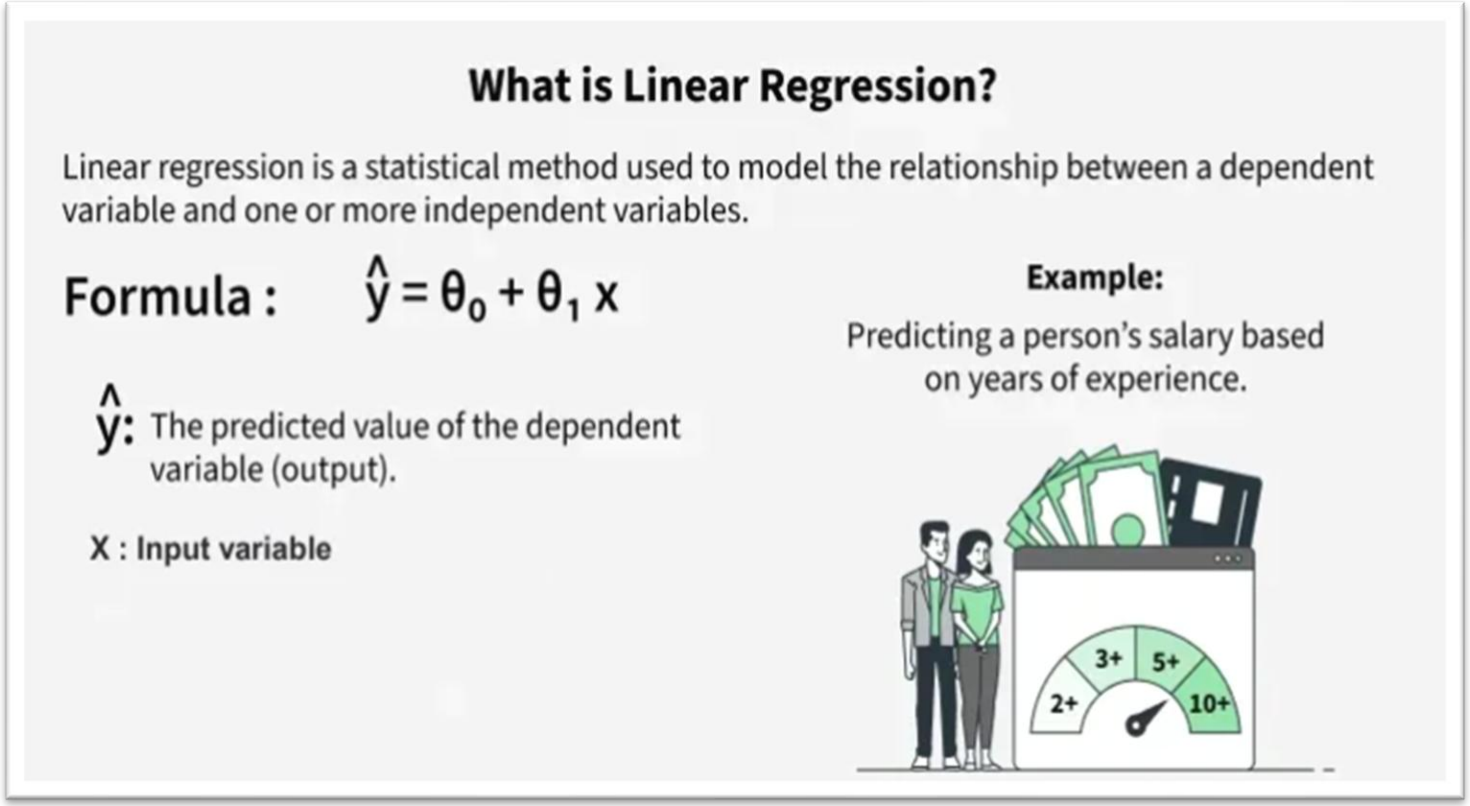
Types of Supervised learning

Regression:

- Regression algorithms are used to solve regression problems in which there is a **linear relationship** between input and output variables.
- These are used to predict continuous output variables, such as **market trends, weather predict**ion, etc.

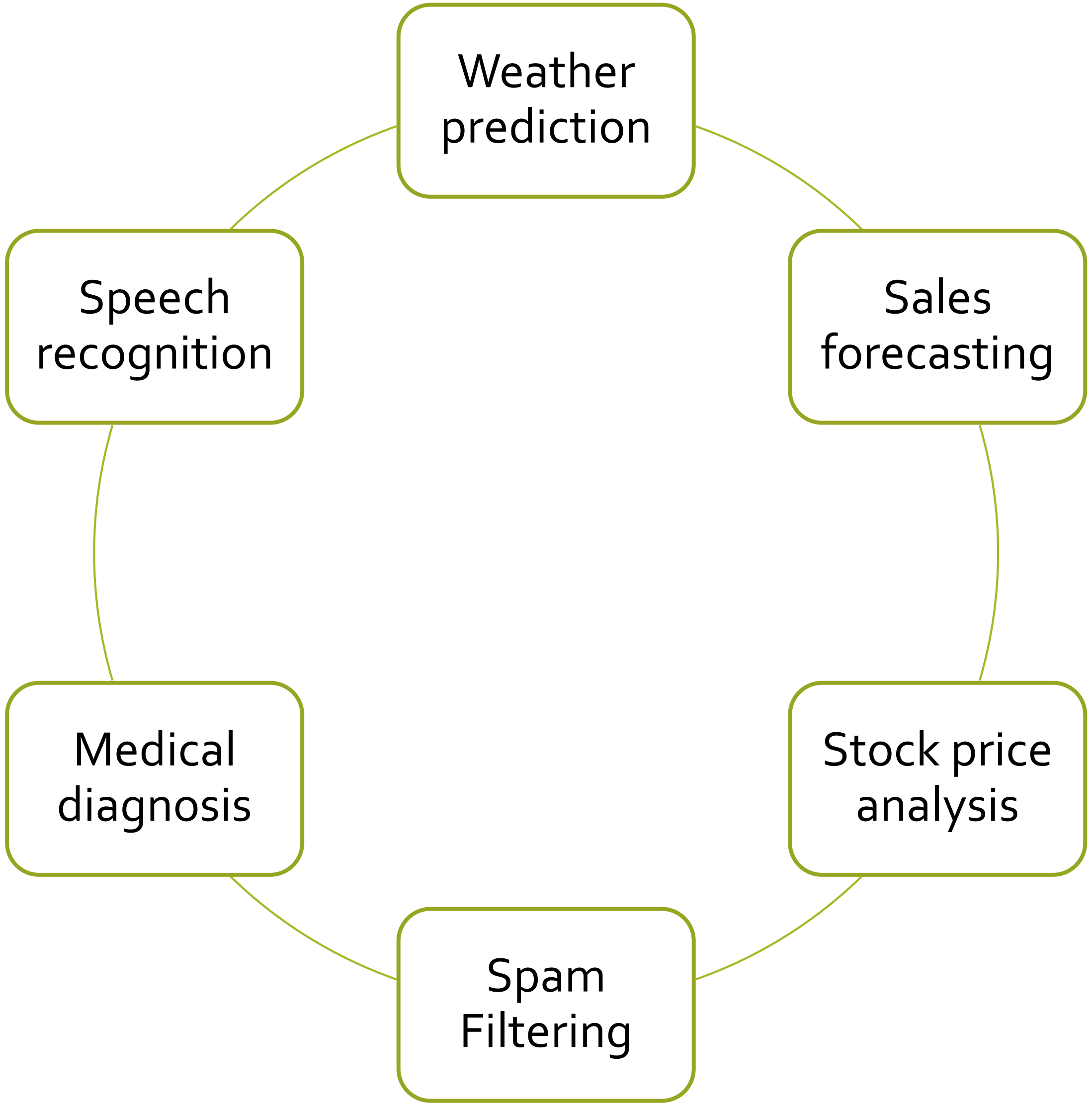
Linear Regression

- Linear regression is one of the simplest machine learning algorithms available, it is used to learn to predict continuous value (dependent variable) based on the features (independent variable) in the training dataset. The value of the dependent variable which represents the effect, is influenced by changes in the value of the independent variable.

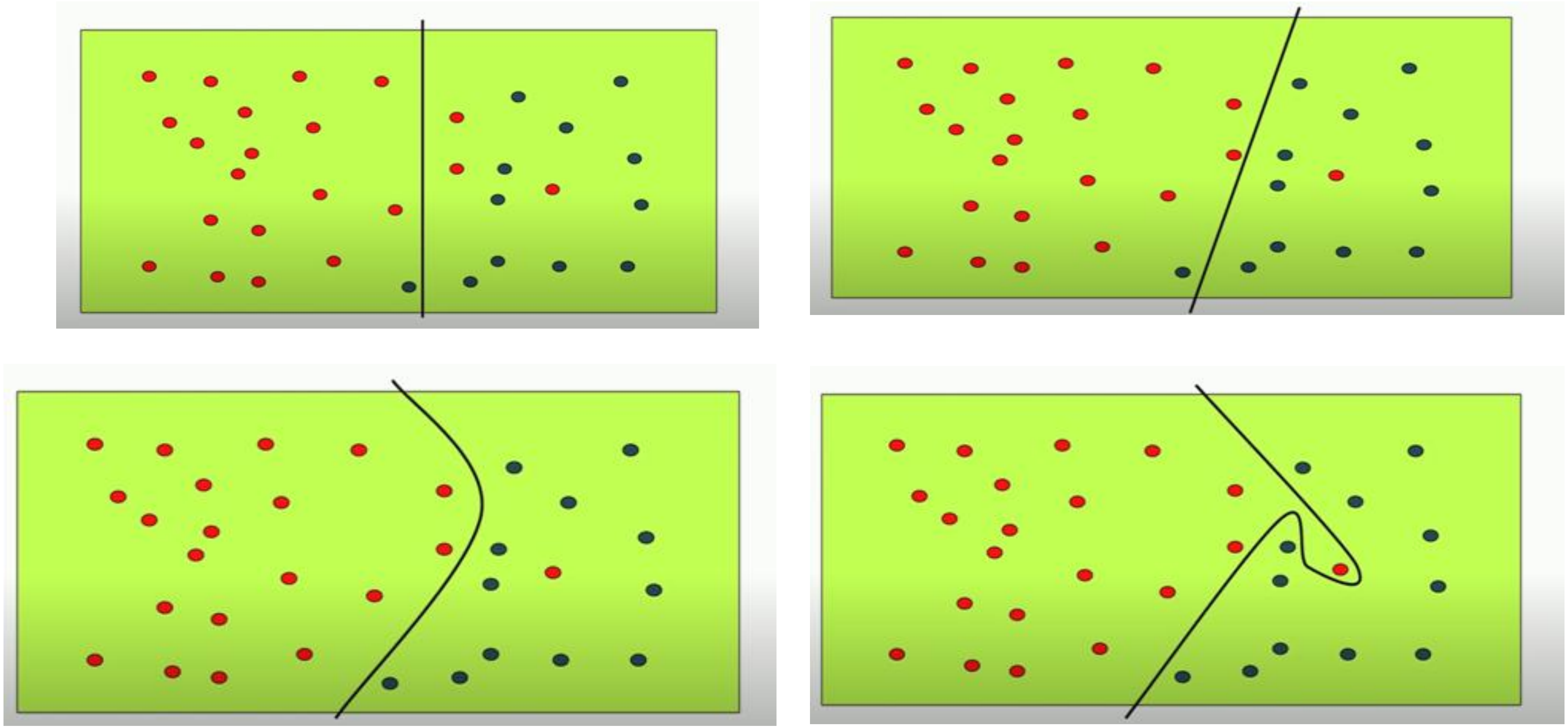


- If you recall the "line of best fit" from school days, this is exactly what linear regression is. Predicting a person's weight based on their height is a straightforward example of this concept.

Supervised learning Applications



Supervised Learning Possible Classifiers



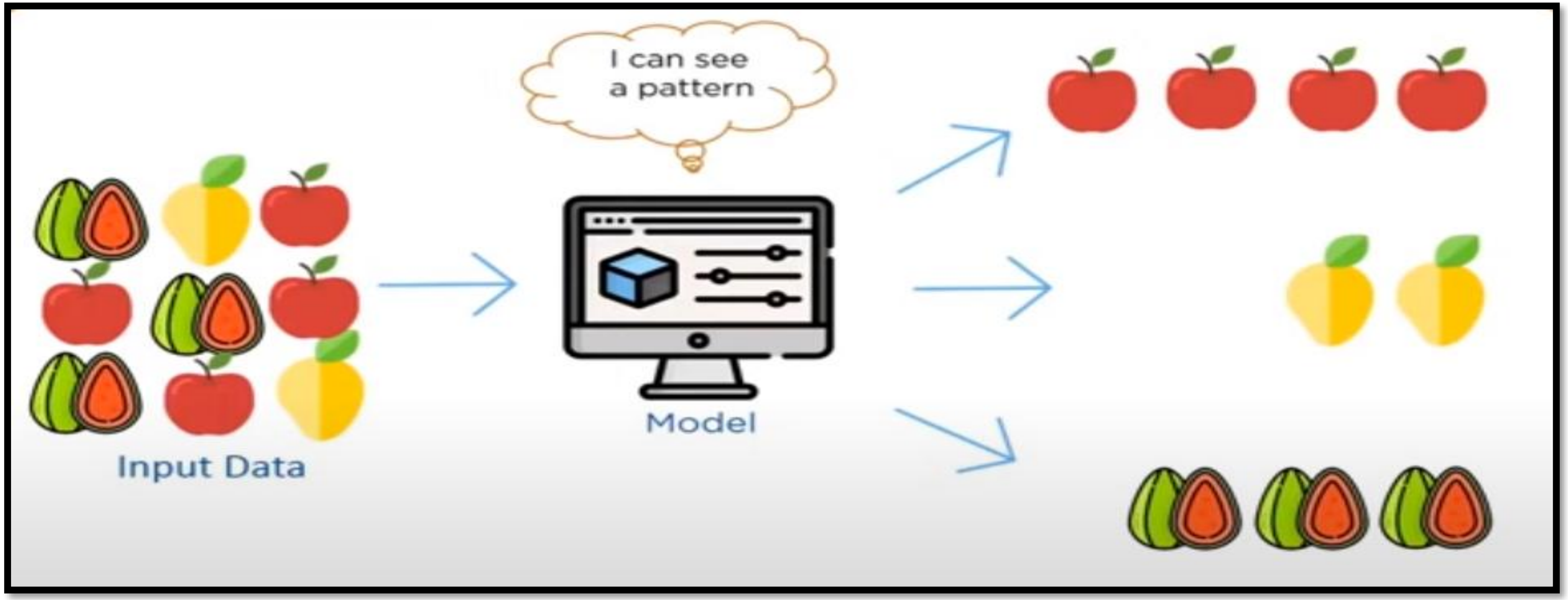
Algorithm	Regression, Classification	Purpose	Method	Use Cases
Linear Regression	Regression	Predict continuous output values	Linear equation minimizing sum of squares of residuals	Predicting continuous values
Logistic Regression	Classification	Predict binary output variable	Logistic function transforming linear relationship	Binary classification tasks
Decision Trees	Both	Model decisions and outcomes	Tree-like structure with decisions and outcomes	Classification and Regression tasks
Random Forests	Both	Improve classification and regression accuracy	Combining multiple decision trees	Reducing overfitting, improving prediction accuracy
SVM	Both	Create hyperplane for classification or predict continuous values	Maximizing margin between classes or predicting continuous values	Classification and Regression tasks
KNN	Both	Predict class or value based on k closest neighbors	Finding k closest neighbors and predicting based on majority or average	Classification and Regression tasks, sensitive to noisy data
Gradient Boosting	Both	Combine weak learners to create strong model	Iteratively correcting errors with new models	Classification and Regression tasks to improve prediction accuracy
Naive Bayes	Classification	Predict class based on feature independence assumption	Bayes' theorem with feature independence assumption	Text classification, spam filtering, sentiment analysis, medical

Unsupervised learning

- It is a learning method in which a machine learns without any supervision.
- Unsupervised learning is a type of machine learning that uses unlabeled data to train machines.
- In unsupervised learning, the models are trained with the data that is neither classified nor labelled, and the model acts on that data without any supervision.
- Unlabeled data doesn’t have a fixed output variable.
- The model learns from the data, discovers the patterns and features in the data, and returns the output.

Unsupervised learning

- **The main aim of the unsupervised learning algorithm is to group or categories the unsorted dataset according to the similarities, patterns, and differences.**
- Machines are instructed to find the hidden patterns from the input dataset.

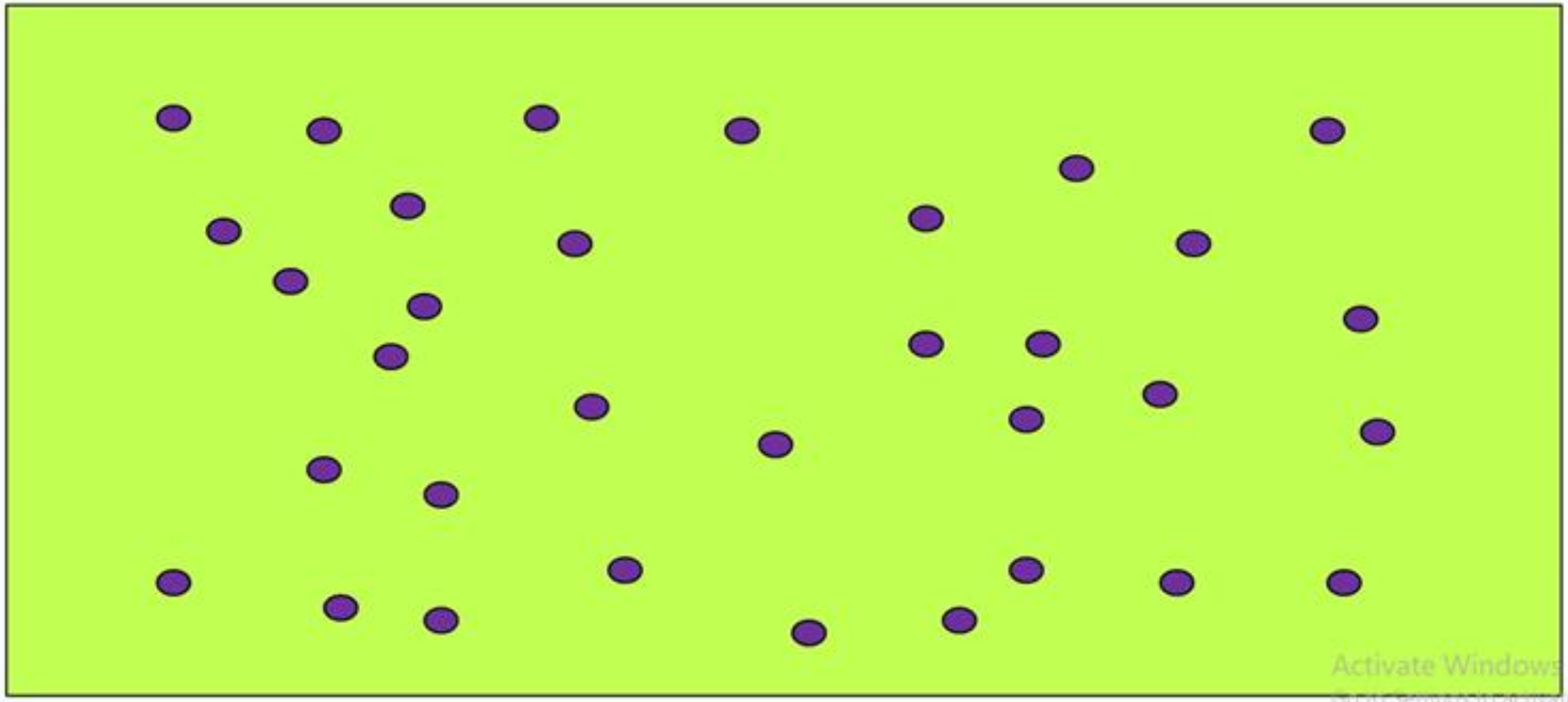


Unsupervised learning

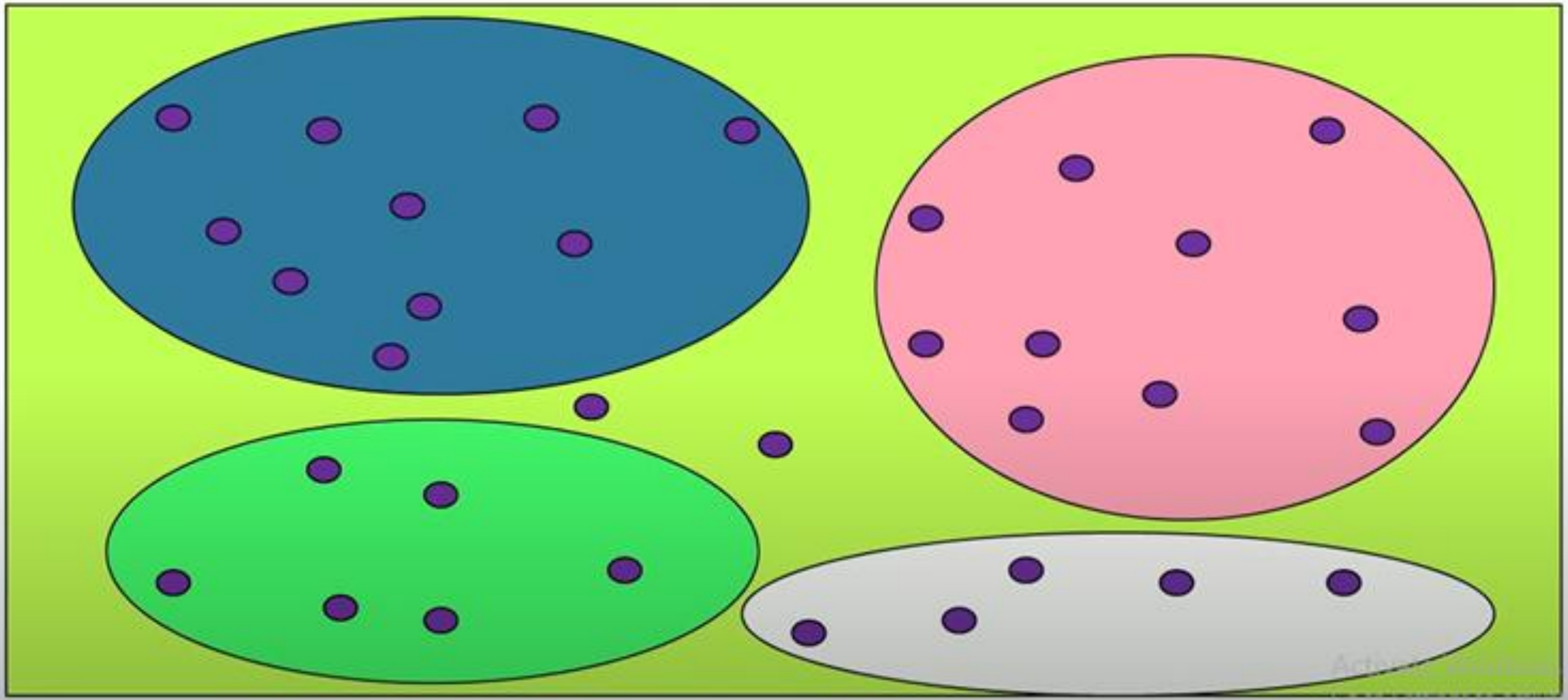
- Unsupervised Learning can be further classified into two types,
  - **Clustering**
  - **Association**

Unlabelled Training Data

Clustering



Possible Clusters



Unsupervised learning

Association

- The clustering technique is used when we want to find the inherent groups from the data.
- It is a way to group the objects into a cluster such that the objects with the most similarities remain in one group and have fewer or no similarities with the objects of other groups.
- An example of the clustering algorithm is grouping the customers by their **purchasing behavior.**

Unsupervised learning

Association

- Association rule learning is an unsupervised learning technique, which finds interesting relations among variables within a large dataset.
- The main aim of this learning algorithm is to find the dependency of one data item on another data item and map those variables accordingly so that it can generate maximum profit.
- This algorithm is mainly applied in **Market Basket analysis, Web usage mining, continuous production**, etc.

Application - Unsupervised learning

- **Customer segmentation:**

  Based on customer behavior, likes, dislikes, and interests, you can segment and **cluster similar customers** into a group.

- **Recommendation Systems:**

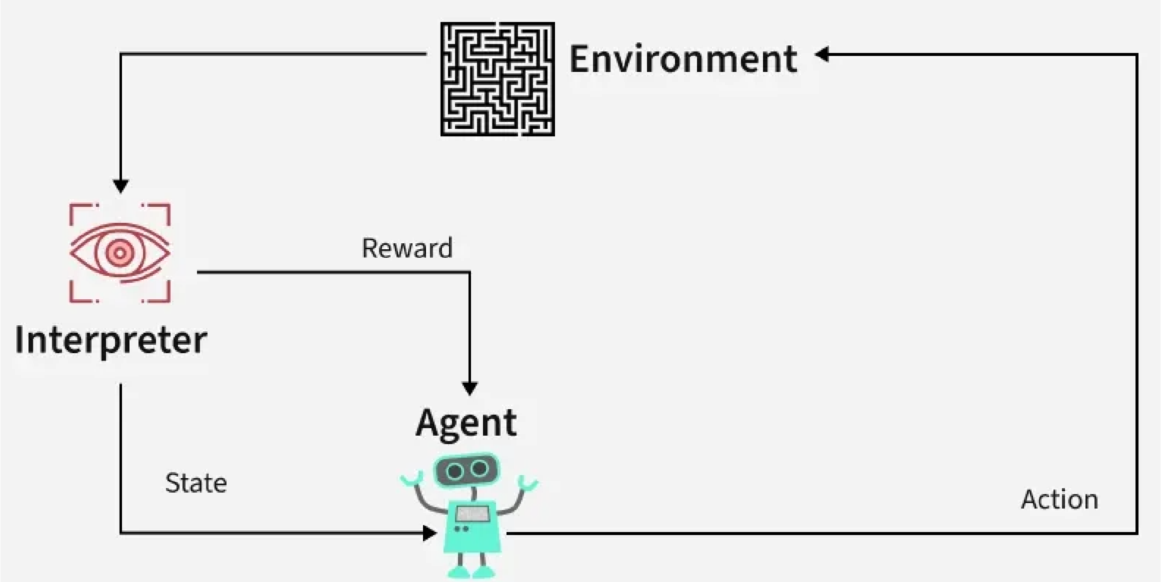
  Recommendation systems widely use unsupervised learning techniques for building recommendation applications for different **web applications and e-commerce websites.**

- **Image and Text Clustering**: Groups similar images or documents for tasks like organization, classification, or content recommendation.

- **Social Network Analysis**: Detects communities or trends in user interactions on social media platforms.

Reinforcement learning

- It is a feedback-based learning method, in which a learning agent gets a reward for each right action and gets a **penalty** for each wrong action.
- Reinforcement learning follows trial and error methods to get the desired result.
- After accomplishing a task, the agent receives an award. An example could be to train a dog to catch the ball.
- If the dog learns to catch a ball, you give it a reward, such as a biscuit.



Applications

- **Robotics:** RL is used to automate tasks in structured environments such as manufacturing, where robots learn to optimize movements and improve efficiency.
- **Game Playing:** Advanced RL algorithms have been used to develop strategies for complex games like chess, Go, and video games, outperforming human players in many instances.
- **Industrial Control:** RL helps in real-time adjustments and optimization of industrial operations, such as refining processes in the oil and gas industry.
- **Personalized Training Systems:** RL enables the customization of instructional content based on an individual's learning patterns, improving engagement and effectiveness.

Popularly used Algorithms

Supervised Learning

- Linear regression
- Logistic regression
- Support Vector Machines
- Naive Bayes
- Decision tree
- Random Forest

Unsupervised Learning

- K-means clustering
- Hierarchical clustering
- DBSCAN.
- Apriori

Reinforcement Learning

- Q-learning
- Deep Q-learning Neural Networks.

Steps in developing ML application

- Data Collection – Quality data
- Data Pre-processing - Preparing/cleaning the Data
- Model Selection - Analyze Data/choose algorithm
- Model Training - Train the Data
- Evaluation - Test Algorithm
- Performance Tuning - Use in Application
- Prediction - Output

Steps in developing ML application

- **1. Data Collection – Quality Data**
  - Gather data from reliable sources (e.g., files, databases, web)
  - Ensure data is relevant and comprehensive
- **2. Data Pre-processing – Preparing / Cleaning the Data**
  - Handle missing values, remove duplicates
  - Normalize and structure the data
- **3. Model Selection – Analyze Data / Choose Algorithm**
  - Understand the problem type (classification, regression, etc.)
  - Select appropriate algorithms (e.g., Linear Regression, Decision Trees)
  - Consider model complexity, interpretability, and resources
- **4. Model Training – Train the Data**
  - Feed cleaned data to the model
  - Adjust internal weights and parameters

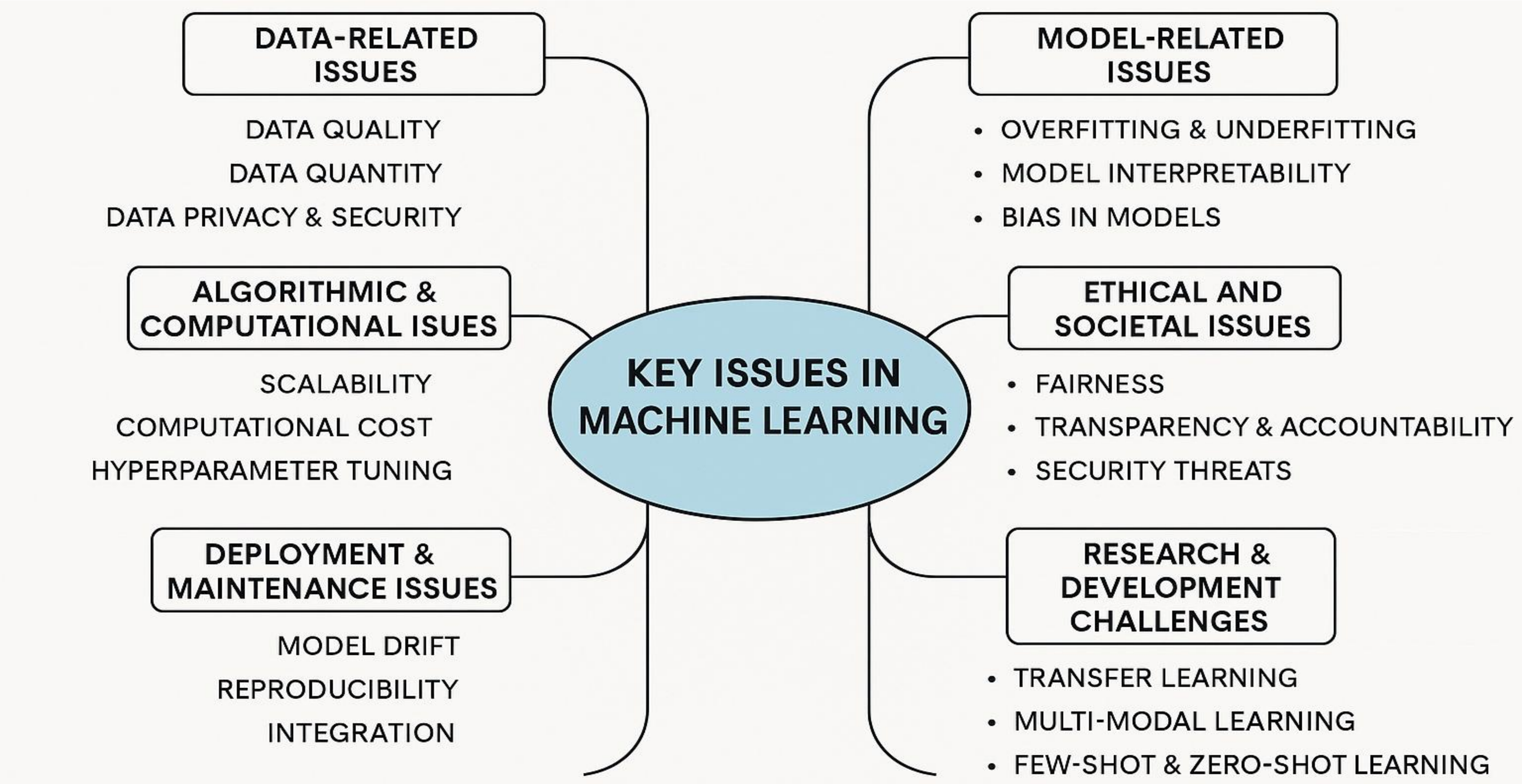
Steps in developing ML application

- **5. Evaluation – *Test the Algorithm***
  - Use test data.
  - Metrics: Accuracy, Precision, Recall, MSE

- **6. Performance Tuning – *Use in Application***
  - Optimize hyperparameters
  - Prevent overfitting/underfitting
  - Prepare for real-world use

- **7. Prediction-** *Generate Output*
  - Deploy the model in a real-world setting
  - Make predictions on new data
  - Integrate into applications or dashboards

Issues of Machine learning



Issues of Machine learning

1. Data-Related Issues

- **Data Quality**: Noisy, incomplete, inconsistent, or biased data can degrade model performance.
- **Data Quantity**: Insufficient training data limits model learning; deep learning models especially require large datasets.
- **Data Privacy & Security**: Using sensitive or personal data raises ethical and legal concerns (e.g., GDPR compliance).
- **Data Labelling**: Supervised learning needs labelled data, which is costly and time-consuming to produce.

2. Model-Related Issues

- **Overfitting & Underfitting**:
  - *Overfitting*: Model learns noise; performs well on training data but poorly on unseen data.
  - *Underfitting*: Model is too simple; fails to capture patterns in training data.
- **Model Interpretability**: Many powerful models (e.g., neural networks) are black boxes, making it hard to understand decisions.
- **Bias in Models**: Models can inherit or amplify biases present in training data, leading to unfair outcomes.
- **Generalization**: Ensuring the model performs well on unseen data or in different domains (domain adaptation).

Issues of Machine learning

3. Algorithmic & Computational Issues

- **Scalability**: Algorithms must efficiently handle large datasets or high-dimensional data.
- **Computational Cost**: Deep learning requires significant resources (GPUs, TPUs, memory).
- **Hyperparameter Tuning**: Model performance can vary greatly depending on tuning, requiring expertise and experimentation.
- **Convergence Issues**: Some algorithms may not converge or may get stuck in local minima.

4. Ethical and Societal Issues

- **Fairness**: Ensuring that ML decisions are unbiased and equitable across all demographics.
- **Transparency & Accountability**: Who is responsible when a model causes harm or makes a mistake?
- **Security Threats**: Susceptibility to adversarial attacks, data poisoning, and model inversion.
- **Explainability**: Growing demand for AI systems to be explainable, especially in sensitive applications (healthcare, law).

Issues of Machine learning

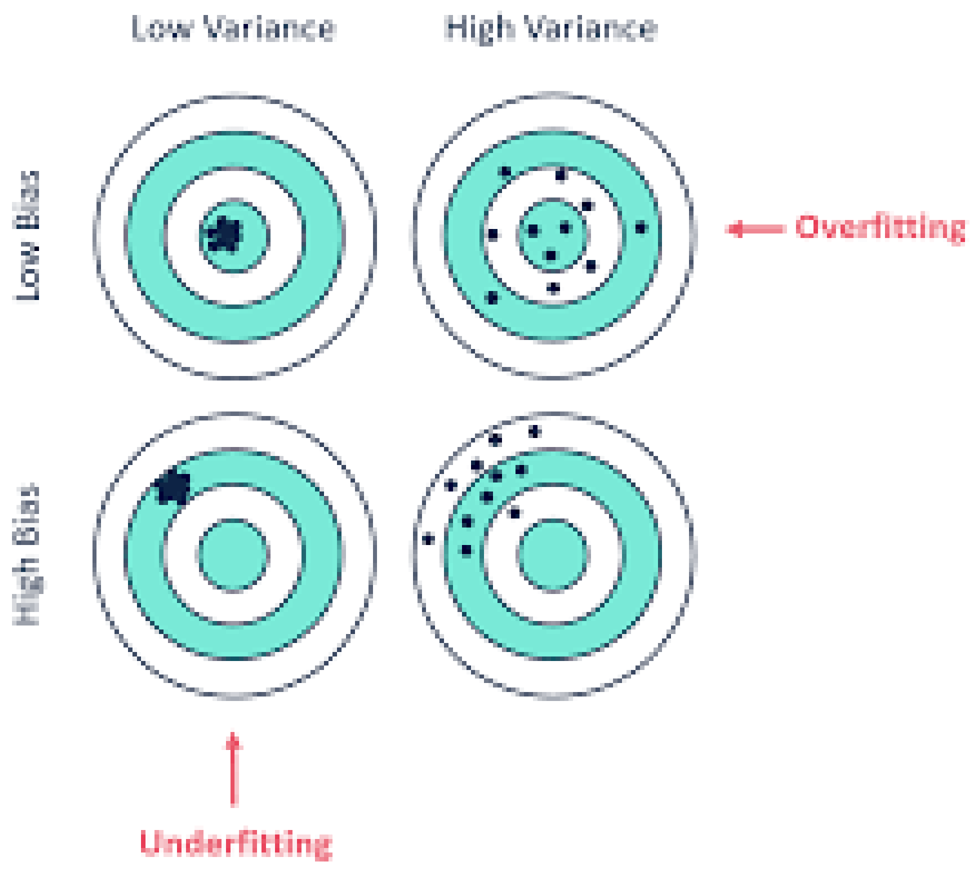
5. Deployment & Maintenance Issues

- **Model Drift**: Data distributions change over time, causing model performance to degrade.
- **Reproducibility**: Difficulty in reproducing results due to random initializations, data splits, or undocumented settings.
- **Integration**: Challenges in deploying models in real-world applications with existing infrastructure.
- **Monitoring**: Need for tools to track performance, errors, and fairness after deployment.

6. Research & Development Challenges

- **Transfer Learning**: Adapting pre-trained models to new tasks with minimal labeled data.
- **Multi-modal Learning**: Integrating data from different sources (e.g., text, image, audio).
- **Few-shot & Zero-shot Learning**: Enabling models to generalize from very few or no examples.
- **Continual Learning**: Teaching models to learn continuously without forgetting previous knowledge.

Bias/ Variance



Bias: Difference in predicted value and actual value

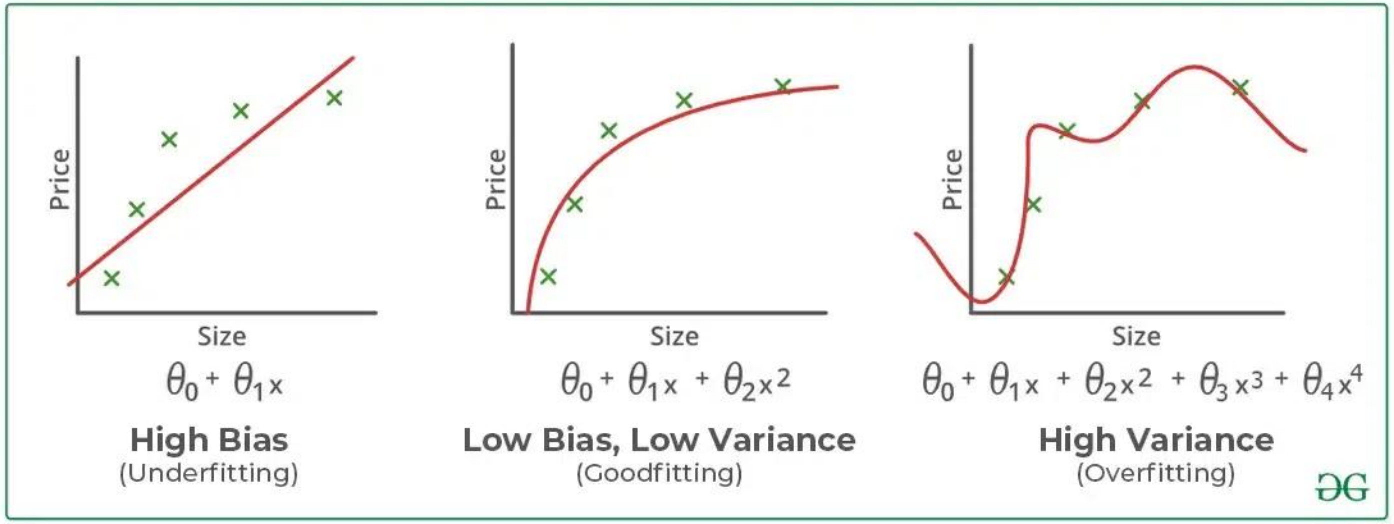
High Bias: difference is more

Variance: how the predicted values are scatter with respect to each other

Low variance: values are not much scattered. They are in groups.

Bias/Variance

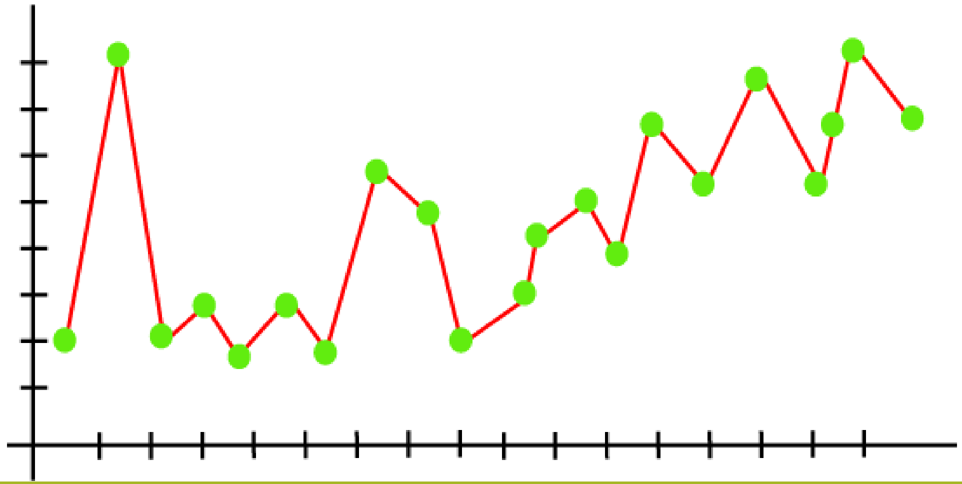
- **Bias:** It is the error due to the model's inability to represent the true relationship between input and output accurately.
- When a model has poor performance both on the training and testing data means high bias because of the simple model, indicating underfitting.
- **Variance:** It is the error due to the model's sensitivity to fluctuations in the training data. It's the variability of the model's predictions for different instances of training data.
- High variance occurs when a model learns the training data's noise and random fluctuations rather than the underlying pattern.
- As a result, the model performs well on the training data but poorly on the testing data, indicating overfitting



Issues in Machine Learning contd..

2. Overfitting of Training Data

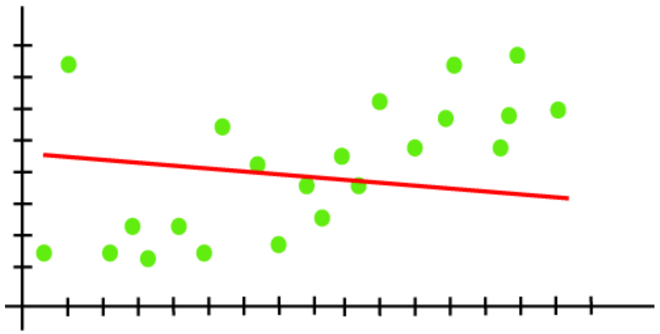
- Whenever a **machine learning model is trained with a huge amount of data**, it starts capturing noise and inaccurate data into the training data set. It negatively affects the performance of the model.
- Overfitting occurs when machine learning model tries to cover all the data points or more than the required data points present in the given dataset.
- The model tries to cover all the data points present in the scatter plot. It may look efficient, but in reality, it is not so. Because the goal of the regression model is to find the best fit line, but here we have not got any best fit, so, it will generate the prediction errors.
- The overfitted model has **low bias** and **high variance.**



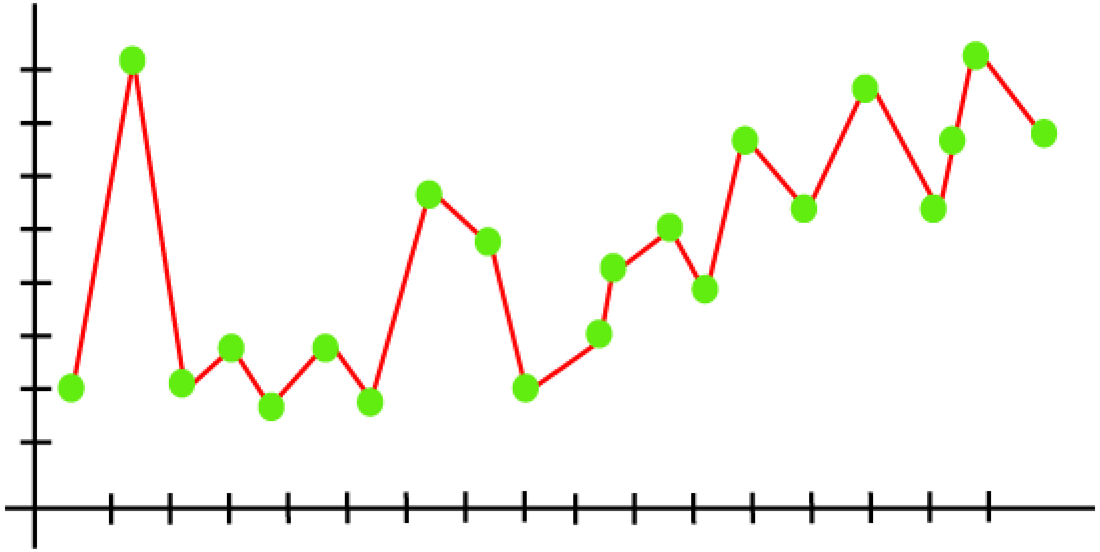
- **Methods to reduce overfitting:**
  - Cross-Validation
  - Training with more data
  - Removing features
  - Early stopping the training
  - Regularization
  - Ensembling

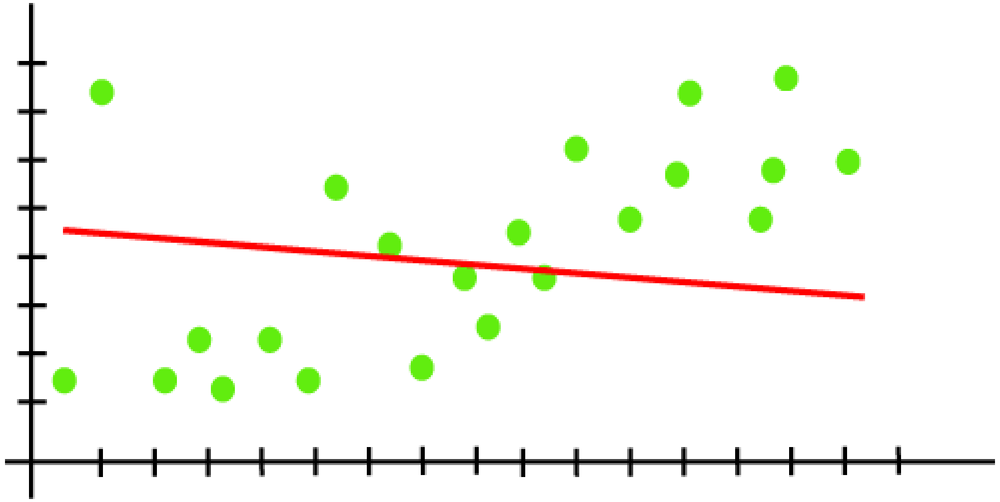
3. Underfitting of Training Data

- Whenever a **machine learning model is trained with fewer amounts of data**, and as a result, it provides incomplete and inaccurate data and destroys the accuracy of the machine learning model.
- This generally happens when we have limited data into the data set, and we try to build a linear model with non-linear data.
- **Methods to reduce Underfitting:**
  - By increasing the training time of the model.
  - By increasing the number of features.



Overfitting & Underfitting





Overfitting: Model with good training set. But may wrong for testing set

Underfitting: Not good for training as well as testing set

Real-life Example of overfitting and underfitting

- **Task:** To identify whether the object is ball or not
- **Parameters:**
- Sphere-This feature is checking if the object is of a spherical shape.
- Play-This feature is checking if one can play with it.
- Eat-This feature is checking if one cannot eat it.
- radius=5 cm-This feature is checking if an object's size is 5 cm or less than it.
- **Overfiiting Case:**
- If object (ball) with 10 cm radius is passed to classifier, it will classify it as not ball. Because classifier is very much specific with features value.
- **Underfitting case:**
- If object (Orange) is passed to classifier , it will classify it as ball. Because it is very much generalized with less number of parameter i.e. if object is sphere in shape it is ball.

Example:

Sphere	play	Eat	Radius	Class
yes	yes	No	5	Ball
yes	yes	No	3	Ball
yes	no	yes	5	Fruit
yes	no	yes	10	Fruit
yes	yes	No	10	Ball

Underfitting: model is buit on two parameters only (Sphere & radius). So when fruit is passed to model it may classify it as ball
Overfitting: Model is specific with all parameters value (**yes,yes,No,5**), so when ball with 10 radius will pass it may classify it as fruit

How to choose a right Algorithm??

- Understand the project goal
- Type of Dataset
- Nature of the problem
- Nature of the Algorithm
- Performance comparison

How to choose a right Algorithm??

Understand the project goal: what kind of an output do you need?

- Do you need an algorithm for prediction based on the previous data?

  Turn to supervised forecasting algorithms.

- Are you looking for an image recognition model that will work with poor-quality photos?

  Dimensionality reduction in combination with classification will help you with it.

- Do you need to teach your model to play a new game?

  A reinforcement algorithm will be your best bet.

How to choose a right Algorithm??

Type of Dataset

- Numerical data set

  SVM, Logistic regression, Decision tree , etc.

- Image Data set

  CNN (Convolutional Neural Network)

- Speech/Text Data Set

  RNN (Recurrent Neural Network)

How to choose a right Algorithm??

Analyze Your Dataset

- What is your data like?
- Is it raw, just collected from wherever, and requires processing?
- Is it biased, dirty, and unstructured?
- Do you have enough data or is additional collecting (or even collecting from scratch) required?
- Do you need to spend time preparing your data for the training process or are you good to go?
  - ✓ Supervised algorithm
  - ✓ Unsupervised algorithm

How to choose a right Algorithm??

Performance Comparison -Evaluate the Speed and Training Time

- Do you need it fast even if it means lower quality of training (and, respectively, predictions)?
- Can you allocate the required time for proper training?

Hypothesis Testing

- Hypothesis Testing is a type of statistical analysis in which you put your assumptions about a population parameter to the test.
- It is used to estimate the relationship between 2 statistical variables.
- Example:
- A doctor believes that 3D (Diet, Dose, and Discipline) is 90% effective for diabetic patients.
- In **machine learning (ML)**, it's used to:
  - Compare model performances.
  - Evaluate feature significance.
  - Validate data assumptions.
  - Make decisions in A/B testing.

How Hypothesis Testing Works?

- An analyst performs hypothesis testing on a statistical sample to present evidence of the plausibility of the null hypothesis.
- Measurements and analyses are conducted on a random sample of the population to test a theory.
- Analysts use a random population sample to test two hypotheses: the null and alternative hypotheses.
- The null hypothesis is typically an equality hypothesis between population parameters; for example, a null hypothesis may claim that the population means return equals zero.
- The alternate hypothesis is essentially the inverse of the null hypothesis (e.g., the population means the return is not equal to zero).
- As a result, they are mutually exclusive, and only one can be correct. One of the two possibilities, however, will always be correct.

2. Key Concepts

- **Null Hypothesis (H_0)**: A statement that there is no effect or no difference, or tha
- **Alternative Hypothesis (H_1 or Ha)**: The opposite of the null hypothesis, indicatin
- **Test Statistic**: A value calculated from the sample data that is used to determine
- **Significance Level (α)**: The probability threshold for rejecting the null hypothesis
- **P-value**: The probability of obtaining a test statistic at least as extreme as the or
- **Type I Error (α)**: Rejecting the null hypothesis when it is actually true (false posit
- **Type II Error (β)**: Failing to reject the null hypothesis when it is actually false (fal

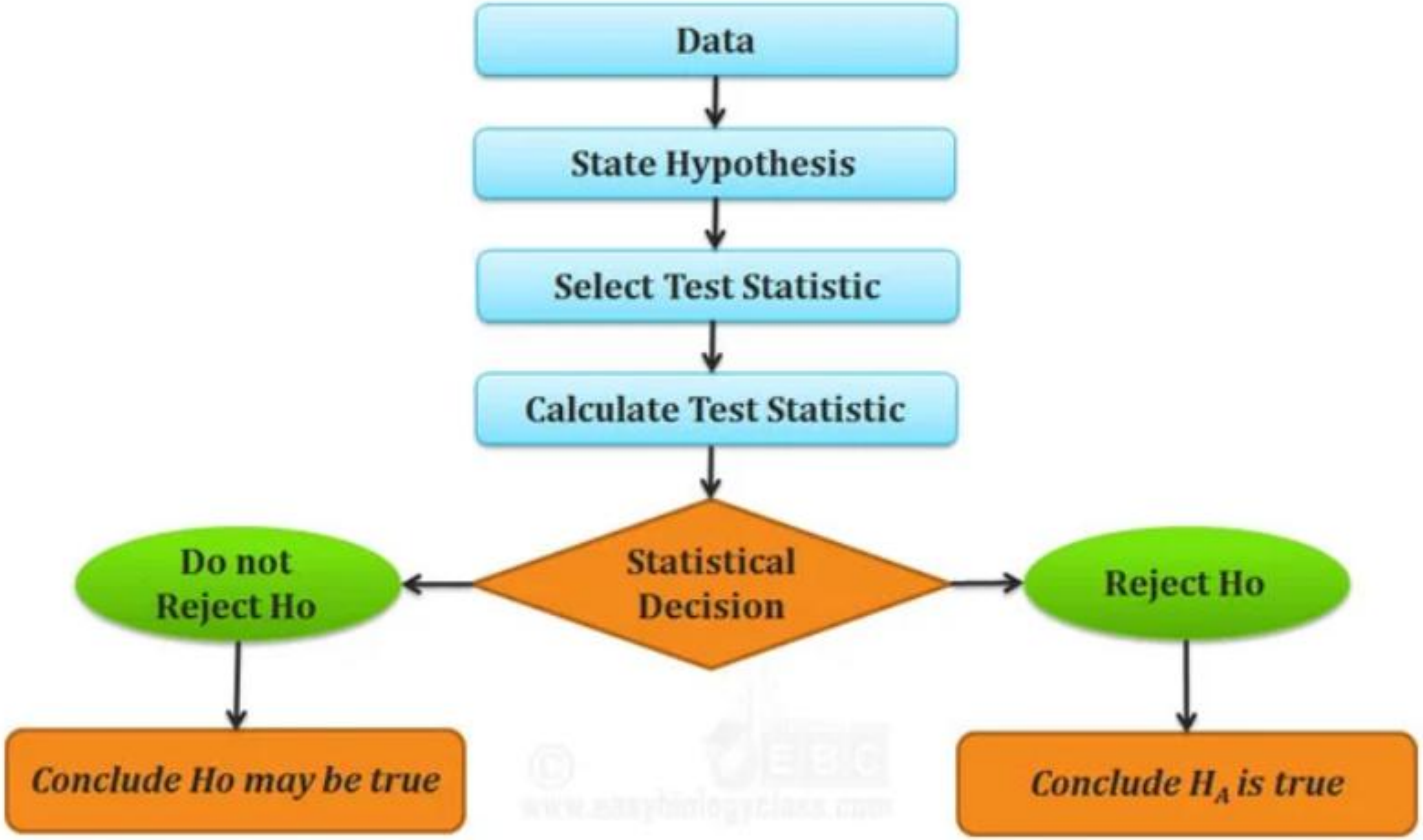
3. Common Hypothesis Tests in ML

Test	Used For	Assumptions
t-test	Compare means of two samples (e.g., model accuracy)	Data follows a normal distribution
z-test	Compare means/proportions (large n, known variance)	Data is normal, σ known
Chi-square test	Independence of categorical variables	Expected freq ≥ 5
ANOVA	Compare >2 group means (e.g., model variants)	Normality, equal variances
Mann-Whitney U test	Non-parametric alternative to t-test	No need for normality

General Procedure for Hypothesis Testing

1. **State the Hypotheses**:
   1. Define the null hypothesis (H_0) and the alternative hypothesis (H_1).
2. **Select the Significance Level (α)**:
   1. Choose the level of significance (commonly $\alpha = 0.05$) which defines the probability of making a Type I error.
3. **Select the Appropriate Test**:
   1. Based on the type of data (e.g., continuous, categorical) and sample size, select the correct statistical test (e.g., Z-test, t-test, chi-square test).
4. **Set the Decision Rule**:
   1. Determine the critical value(s) or the rejection region(s) based on the selected test and significance level.
5. **Compute the Test Statistic**:
   1. Using the sample data, calculate the test statistic (e.g., z, t, chi-square).
6. **Make a Decision**:
   1. Compare the computed test statistic with the critical value or use the p-value to decide whether to reject the null hypothesis.
      1. If the test statistic falls in the rejection region (or $p\text{-value} < \alpha$), reject H_0.
      2. If the test statistic falls in the acceptance region (or $p\text{-value} \geq \alpha$), fail to reject H_0.
7. **Draw a Conclusion**:
   1. Based on the decision, conclude whether there is sufficient evidence to support the alternative hypothesis.

STEPS IN HYPOTHESIS TESTING



Types of Hypothesis Testing

1. Z-Test

- **When to Use:**
- Large sample size (**n > 30**)
- Population standard deviation (σ) is **known**
- Data is **normally distributed** or approximately normal

- **Types of Z-Test:**
  - **One-sample Z-test**: Tests if a sample mean is significantly different from a known population mean.
  - **Two-sample Z-test**: Compares the means of two independent groups.

- **Application in ML:**
  - Comparing performance (e.g., accuracy) of two models when sample size is large.

Types of Hypothesis Testing

2. T-Test

- **When to Use:**
- Small sample size (**n < 30**)
- Population standard deviation is **unknown**
- Data should be **approximately normal**

- **Application in ML:**
  - Cross-validation: Compare model A vs model B performance using paired t-test.

- **Types:**

Test Type	Use Case
One-sample t-test	Compare sample mean to a known value
Two-sample t-test (independent)	Compare two independent samples (e.g., accuracy of two different models)
Paired t-test (dependent)	Compare two related samples (e.g., accuracy of same model on two datasets)

Types of Hypothesis Testing

3. Chi-Square Test (χ^2 Test)

- **When to Use:**
- For **categorical variables**
- To test relationships between variables or expected vs observed frequencies

- **Application in ML:**
  - Feature selection (e.g., selecting relevant categorical features for classification)
  - Assessing distributions in data preprocessing

- Types:

Test Type	Purpose
Test of Independence	Is there a relationship between two categorical variables?
Goodness-of-Fit Test	Does observed distribution match expected distribution?

Types of Hypothesis Testing

4. ANOVA (Analysis of Variance)

- **When to Use:**
- Comparing **more than two groups** (means)
- Checks if at least one group mean is different

- **Application in ML:**
  - Comparing performance across **multiple models** or **algorithms**
  - Testing the impact of **hyperparameters** or experimental settings

- **Types:**

Type	Use Case
One-Way ANOVA	One independent variable (e.g., testing accuracy across 3 models)
Two-Way ANOVA	Two independent variables and their interaction effects

Test Type	Purpose	Test Statistic Formula	Assumptions
Z-Test	Test population mean when σ is known.	$Z = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}$	- Normally distributed data or large sample ($n \geq 30$). - Known population SD (σ).
One-Sample t-Test	Test population mean when σ is unknown.	$t = \frac{\bar{X} - \mu_0}{s / \sqrt{n}}$	- Normally distributed data or large sample. - Unknown population SD.
Two-Sample t-Test	Compare means of two independent groups.	$t = \frac{(\bar{X}_1 - \bar{X}_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$	- Independent samples. - Normally distributed or large samples. - Equal variances (if pooled).
Paired t-Test	Compare means of two related groups (before/after).	$t = \frac{\bar{D} - \mu_D}{s_D / \sqrt{n}}$ (where $D = X_1 - X_2$)	- Paired observations. - Normally distributed differences.
Chi-Square Goodness-of-Fit	Test if sample matches expected distribution.	$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$	- Categorical data. - Large enough expected frequencies (≥5 per category).
Chi-Square Test of Independence	Test association between two categorical variables.	$\chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$	- Independent observations. - Large enough expected frequencies.
ANOVA (F-Test)	Compare means of three or more groups.	$F = \frac{\text{Between-group variance}}{\text{Within-group variance}}$	- Normally distributed data. - Homogeneity of variances. - Independent groups.

Why is Hypothesis Testing Important in ML?

- **Provides evidence-based conclusions**: It allows researchers to make objective conclusions based on empirical data, providing evidence to support or refute their research hypotheses.
- **Supports decision-making**: It helps make informed decisions, such as accepting or rejecting a new treatment, implementing policy changes, or adopting new practices.
- **Adds rigor and validity**: It adds scientific rigor to research using statistical methods to analyse data, ensuring that conclusions are based on sound statistical evidence.
- **Contributes to the advancement of knowledge**: By testing hypotheses, researchers contribute to the growth of knowledge in their respective fields by confirming existing theories or discovering new patterns and relationships.

Limitations of Hypothesis Testing

- **It cannot prove or establish the truth**: Hypothesis testing provides evidence to support or reject a hypothesis, but it cannot confirm the absolute truth of the research question.
- **Results are sample-specific**: Hypothesis testing is based on analyzing a sample from a population, and the conclusions drawn are specific to that particular sample.
- **Possible errors**: During hypothesis testing, there is a chance of committing type I error (rejecting a true null hypothesis) or type II error (failing to reject a false null hypothesis).
- **Assumptions and requirements**: Different tests have specific assumptions and requirements that must be met to accurately interpret results.

- https://www.appliedaicourse.com/blog/issues-in-machine-learning/